SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

Kaviya k

I (192521262) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

k.kaviya

Q1. Probability of Selecting Box C Given a Blue Ball

Given:

Box Red Blue Green Total

A 6 4 2 12

B 5 7 3 15

C 8 2 5 15

Box B is twice as likely to be selected as A or C.

Step 1: Calculate prior probabilities

$$ P(A)=\frac14,\quad P(B)=\frac12,\quad P(C)=\frac14 $$

Step 2: Calculate blue-ball probabilities

$$ P(Blue|A)=\frac4{12}=\frac13 $$

$$ P(Blue|B)=\frac7{15} $$

$$ P(Blue|C)=\frac2{15} $$

Step 3: Apply Bayes' Theorem

$$ P(C|Blue)=\frac{P(Blue|C)P(C)}{P(Blue)} $$

$$ =\frac{\frac2{15}\times\frac14}
{\frac13\times\frac14+\frac7{15}\times\frac12+\frac2{15}\times\frac14} $$

$$ =\frac{\frac1{30}}{\frac7{20}} $$

$$ =\frac2{21} $$

Final Answer:

$$ \boxed{P(C|Blue)=0.0952=9.52\%} $$

Q2. Probability of Selecting Section C Given a Blue Car

Given:

Section Red Blue Black Total

A 5 3 2 10

B	4	6	5	15
C	7	2	6	15

Step 1: Prior probabilities

$$P(A)=\frac14,\quad P(B)=\frac12,\quad P(C)=\frac14$$

Step 2: Likelihood probabilities

$$P(Blue|A)=\frac3{10}$$

$$P(Blue|B)=\frac6{15}=\frac25$$

$$P(Blue|C)=\frac2{15}$$

Step 3: Apply Bayes' Theorem

$$P(C|Blue)= \frac{P(Blue|C)P(C)} {\sum P(Blue|Section)P(Section)}$$

$$=\frac{\frac2{15}\times\frac14}{\frac3{10}\times\frac14+\frac25\times\frac12+\frac2{15}\times\frac14}$$

$$=\frac{\frac1{30}}{\frac{37}{120}}$$

$$=\frac4{37}$$

Final Answer:

$$\boxed{P(C|Blue)=10.81\%}$$

Q3. Breast Cancer Classification – Confusion Matrix

Given:

- Total dataset = 1800 + 1800 = 3600
- Testing data = 20%
- TP = 210
- TN = 190
- FP = 30

Step 1: Calculate testing samples

$$Testing=3600\times0.20=720$$

Step 2: Calculate False Negatives

$$ FN=Total-(TP+TN+FP) $$

$$ FN=720-(210+190+30) $$

$$ \boxed{FN=290} $$

Step 3: Calculate performance metrics

1. Accuracy

$$ Accuracy=\frac{TP+TN}{TP+TN+FP+FN}\times100 $$

$$ =\frac{210+190}{720}\times100 $$

$$ \boxed{55.56\%} $$

2. Precision

$$ Precision=\frac{TP}{TP+FP}\times100 $$

$$ =\frac{210}{210+30}\times100 $$

$$ \boxed{87.50\%} $$

3. Sensitivity (Recall)

$$ Recall=\frac{TP}{TP+FN}\times100 $$

$$ =\frac{210}{210+290}\times100 $$

$$ \boxed{42\%} $$

4. Specificity

$$ Specificity=\frac{TN}{TN+FP}\times100 $$

$$ =\frac{190}{190+30}\times100 $$

$$ \boxed{86.36\%} $$

Conclusion

The model has high precision but relatively low sensitivity. This indicates that it identifies many predicted positive cases correctly but misses a substantial number of actual positive cases. In medical diagnosis, improving sensitivity is particularly important.

Note: The question's stated class distribution implies 360 positive and 360 negative test samples. The supplied TP, TN, and FP values do not match those class totals. The calculation above uses the total test size of 720 to determine FN.

Q4. Training Loss and Validation Loss – Overfitting

(i) Identify the epoch at which overfitting begins.

Validation loss decreases until Epoch 3, reaching 2.0. It then increases from Epoch 4 while training loss continues decreasing.

Answer: Overfitting begins at Epoch 4.

(ii) Suggest two techniques to reduce overfitting.

1. Dropout: Randomly disables selected neurons during training to reduce dependence on particular neurons.
2. Early Stopping: Stops training when validation loss begins increasing to prevent unnecessary learning from training data.

(iii) How does overfitting affect generalization?

Overfitting occurs when a model learns training data too closely, including noise and unnecessary patterns. Consequently, training performance improves while performance on unseen data deteriorates.

This reduces the model's generalization ability.

Q5. Training Accuracy and Validation Accuracy

(i) Identify the epoch at which overfitting starts.

Validation accuracy reaches its highest value of 73% at Epoch 5 and decreases afterward, even though training accuracy continues increasing.

Answer: Overfitting becomes evident at Epoch 6.

(ii) Why does validation accuracy decrease?

The model increasingly memorizes the training data rather than learning general patterns. As a result, it performs poorly on unseen validation samples.

(iii) Suggest two methods to improve validation performance.

1. Data Augmentation: Increases training-data variety through transformations such as rotation, flipping, or cropping.
2. Regularization: Techniques such as L1 or L2 regularization discourage excessive model complexity and help improve generalization.

Q6. Training and Validation Loss Analysis

(i) At which epoch does the model achieve optimal generalization?

The minimum validation loss is 2.0 at Epoch 4.

$$\boxed{\text{Optimal Epoch}=4}$$

(ii) Identify signs of overfitting.

- Training loss continuously decreases.
- Validation loss decreases until Epoch 4.
- Validation loss increases from Epoch 5 onward.
- The increasing gap between training and validation loss indicates overfitting.

(iii) Recommend two regularization techniques.

1. L2 Regularization: Adds a penalty for large model weights, reducing model complexity.
2. Dropout: Randomly deactivates neurons during training, preventing excessive dependence on specific features.

Q7. Industrial Tire Classification

Given:

- TP = 119
- FN = 6
- TN = 120
- FP = 5
- Total = 250

1. Accuracy

$$=\frac{TP+TN}{TP+TN+FP+FN}\times100$$

$$=\frac{119+120}{250}\times100$$

$$\boxed{95.60\%}$$

2. Precision

$$=\frac{TP}{TP+FP}\times100$$

$$=\frac{119}{119+5}\times100$$

$$ \boxed{95.97\%} $$

3. Sensitivity (Recall)

$$ =\frac{TP}{TP+FN}\times100 $$

$$ =\frac{119}{119+6}\times100 $$

$$ \boxed{95.20\%} $$

4. Specificity

$$ =\frac{TN}{TN+FP}\times100 $$

$$ =\frac{120}{120+5}\times100 $$

$$ \boxed{96\%} $$

5. F1-Score

$$ F1=\frac{2TP}{2TP+FP+FN} $$

$$ =\frac{238}{238+5+6}\times100 $$

$$ \boxed{95.58\%} $$

Q8. Medical Disease Classification

Given:

- TP = 138
- FN = 12
- TN = 135
- FP = 15
- Total = 300

1. Accuracy

$$ =\frac{138+135}{300}\times100 $$

$$ \boxed{91\%} $$

2. Precision

$$ =\frac{138}{138+15}\times100 $$

$$ \boxed{90.20\%} $$

3. Sensitivity (Recall)

$$=\frac{138}{138+12}\times100$$

$$\boxed{92\%}$$

4. Specificity

$$=\frac{135}{135+15}\times100$$

$$\boxed{90\%}$$

5. F1-Score

$$=\frac{2(138)}{2(138)+15+12}\times100$$

$$=\frac{276}{303}\times100$$

$$\boxed{91.09\%}$$

Q9. Spam Email Classification

Given:

- TP = 180
- FN = 20
- TN = 170
- FP = 30
- Total = 400

1. Accuracy

$$=\frac{180+170}{400}\times100$$

$$\boxed{87.50\%}$$

2. Precision

$$=\frac{180}{180+30}\times100$$

$$\boxed{85.71\%}$$

3. Sensitivity (Recall)

$$=\frac{180}{180+20}\times100$$

$$\boxed{90\%}$$

4. Specificity

$$ =\frac{170}{170+30}\times100 $$

$$ \boxed{85\%} $$

5. F1-Score

$$ =\frac{2(180)}{2(180)+30+20}\times100 $$

$$ =\frac{360}{410}\times100 $$

$$ \boxed{87.80\%} $$

Q10. Manufacturing Quality Inspection

Given:

- TP = 230
- FN = 20
- TN = 225
- FP = 25
- Total = 500

The question's TP value is printed as "2 30"; this solution interprets it as 230.

1. Accuracy

$$ =\frac{230+225}{500}\times100 $$

$$ \boxed{91\%} $$

2. Precision

$$ =\frac{230}{230+25}\times100 $$

$$ \boxed{90.20\%} $$

3. Sensitivity

$$ =\frac{230}{230+20}\times100 $$

$$ \boxed{92\%} $$

4. Specificity

$$ =\frac{225}{225+25}\times100 $$

$$\boxed{90\%}$$

5. F1-Score

$$=\frac{2(230)}{2(230)+25+20}\times100$$

$$=\frac{460}{505}\times100$$

$$\boxed{91.09\%}$$

Q11. Probability of Adulterated Oil from Shop X or Shop Y

Given:

Shop	Pure Oil	Adulterated Oil	Total
X	20	30	50
Y	40	20	60

Shop Y is twice as likely to be selected as Shop X.

Step 1: Prior probabilities

$$P(X)=\frac13,\quad P(Y)=\frac23$$

Step 2: Likelihood probabilities

$$P(A|X)=\frac{30}{50}=0.6$$

$$P(A|Y)=\frac{20}{60}=\frac13$$

Here, A represents adulterated oil.

Step 3: Apply Bayes' Theorem

Probability that the adulterated bottle came from Shop X:

$$P(X|A)=\frac{P(A|X)P(X)}{P(A|X)P(X)+P(A|Y)P(Y)}$$

$$=\frac{0.6\times\frac13}{0.6\times\frac13+\frac13\times\frac23}$$

$$=\frac9{19}$$

$$\boxed{P(X|A)=47.37\%}$$

Probability that the adulterated bottle came from Shop Y:

$$P(Y|A)=\frac{\frac13\times\frac23}{0.6\times\frac13+\frac13\times\frac23}$$

$$=\frac{10}{19}$$

$$\boxed{P(Y|A)=52.63\%}$$

Q12. Probability of Selecting a Defective Item from Section A or B

Given:

Section	Defective	Non-Defective	Total
A	15	35	50
B	25	25	50

Section A is three times as likely to be selected as Section B.

Step 1: Prior probabilities

$$P(A)=\frac34,\quad P(B)=\frac14$$

Step 2: Likelihood probabilities

$$P(D|A)=\frac{15}{50}=0.3$$

$$P(D|B)=\frac{25}{50}=0.5$$

Step 3: Apply Bayes' Theorem

Probability that the defective item came from Section A:

$$P(A|D)=\frac{0.3\times0.75} {0.3\times0.75+0.5\times0.25}$$

$$=\frac9{14}$$

$$\boxed{P(A|D)=64.29\%}$$

Probability that the defective item came from Section B:

$$P(B|D)=\frac{0.5\times0.25} {0.3\times0.75+0.5\times0.25}$$

$$=\frac5{14}$$

$$\boxed{P(B|D)=35.71\%}$$

Q13. Deep Learning with KNN – Classification Metrics

Given:

- TP = 120
- TN = 150
- Total samples = 1000 × 2 = 2000

- Testing percentage = 20%

Step 1: Calculate testing samples

$$ 2000\times0.20=400 $$

Each class contributes 200 testing samples.

$$ FN=200-120=80 $$

$$ FP=200-150=50 $$

1. Accuracy

$$ =\frac{TP+TN}{TP+TN+FP+FN}\times100 $$

$$ =\frac{120+150}{400}\times100 $$

$$ \boxed{67.50\%} $$

2. Precision

$$ =\frac{120}{120+50}\times100 $$

$$ \boxed{70.59\%} $$

3. Sensitivity (Recall)

$$ =\frac{120}{120+80}\times100 $$

$$ \boxed{60\%} $$

4. Specificity

$$ =\frac{150}{150+50}\times100 $$

$$ \boxed{75\%} $$

Q14. Medical Image Classification Using KNN

Given:

- TP = 180
- TN = 160
- Total dataset = 1200 × 2 = 2400
- Testing percentage = 25%

Step 1: Calculate testing samples

$$ 2400\times0.25=600 $$

Each class contributes 300 testing samples.

$$ FN=300-180=120 $$

$$ FP=300-160=140 $$

1. Accuracy

$$ =\frac{180+160}{600}\times100 $$

$$ \boxed{56.67\%} $$

2. Precision

$$ =\frac{180}{180+140}\times100 $$

$$ \boxed{56.25\%} $$

3. Sensitivity

$$ =\frac{180}{180+120}\times100 $$

$$ \boxed{60\%} $$

4. Specificity

$$ =\frac{160}{160+140}\times100 $$

$$ \boxed{53.33\%} $$

Q15. Training and Validation Loss – Generalization

(a) At which epoch does the model achieve the best generalization performance?

The minimum validation loss is 1.8 at Epoch 4.

$$ \boxed{\text{Best Generalization}=Epoch\ 4} $$

(b) At which point does overfitting begin? Justify your answer.

Overfitting begins at Epoch 5.

Justification:

- Training loss continues decreasing from 1.9 to 1.7.
- Validation loss increases from 1.8 to 1.9.
- This indicates that the model is learning the training data while its performance on unseen data begins deteriorating.

(c) Suggest two techniques to reduce overfitting.

1. Early Stopping

Monitors validation loss and stops training when validation performance begins deteriorating. This prevents unnecessary training.

2. Dropout

Randomly disables selected neurons during training, reducing dependence on specific neurons and improving generalization.

Quick Revision – Final Answers

Q.No.	Final Answer
1	9.52%
2	10.81%
3	FN = 290; Accuracy = 55.56%; Precision = 87.50%; Recall = 42%; Specificity = 86.36%
4	Overfitting begins at Epoch 4
5	Overfitting evident at Epoch 6
6	Optimal generalization at Epoch 4
7	Accuracy 95.60%; Precision 95.97%; Recall 95.20%; Specificity 96%; F1 95.58%
8	Accuracy 91%; Precision 90.20%; Recall 92%; Specificity 90%; F1 91.09%
9	Accuracy 87.50%; Precision 85.71%; Recall 90%; Specificity 85%; F1 87.80%
10	Accuracy 91%; Precision 90.20%; Recall 92%; Specificity 90%; F1 91.09%
11	Shop X 47.37%; Shop Y 52.63%
12	Section A 64.29%; Section B 35.71%
13	Accuracy 67.50%; Precision 70.59%; Recall 60%; Specificity 75%
14	Accuracy 56.67%; Precision 56.25%; Recall 60%; Specificity 53.33%
15	Best generalization Epoch 4; overfitting begins Epoch 5

Give feedback

no step name needed am just going to copy paste so give qs and answers only

MACHINE LEARNING ASSIGNMENT – CO1 AT1

Course Code: ITA 06 | Course Name: Machine Learning

Q1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

Answer:

Given,

P(A) = 1/4, P(B) = 1/2, P(C) = 1/4

P(Blue|A) = 4/12 = 1/3

P(Blue|B) = 7/15

P(Blue|C) = 2/15

Using Bayes' Theorem,

P(C|Blue) = [P(Blue|C) × P(C)] / P(Blue)

= [(2/15) × (1/4)] / [(1/3 × 1/4) + (7/15 × 1/2) + (2/15 × 1/4)]

= (1/30) / (7/20)

= 2/21

Final Answer: 9.52%

Q2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

Answer:

Given,

P(A) = 1/4, P(B) = 1/2, P(C) = 1/4

P(Blue|A) = 3/10

P(Blue|B) = 6/15 = 2/5

P(Blue|C) = 2/15

Using Bayes' Theorem,

P(C|Blue) = [P(Blue|C) × P(C)] / P(Blue)

= [(2/15) × (1/4)] / [(3/10 × 1/4) + (2/5 × 1/2) + (2/15 × 1/4)]

= (1/30) / (37/120)

= 4/37

Final Answer: 10.81%

Q3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Determine the missing False Negatives (FN) value and calculate Accuracy, Precision, Sensitivity (Recall), and Specificity.

Answer:

Total Samples = 1800 + 1800 = 3600

Testing Samples = 3600 × 20/100 = 720

TP = 210, TN = 190, FP = 30

FN = 720 − (210 + 190 + 30)

FN = 290

Accuracy

= (TP + TN) / (TP + TN + FP + FN) × 100

= (210 + 190) / 720 × 100

= 55.56%

Precision

= TP / (TP + FP) × 100

= 210 / (210 + 30) × 100

= 87.50%

Sensitivity (Recall)

= TP / (TP + FN) × 100

= 210 / (210 + 290) × 100

= 42%

Specificity

= TN / (TN + FP) × 100

= 190 / (190 + 30) × 100

= 86.36%

The model has high precision but relatively low sensitivity. This indicates that it correctly identifies many predicted positive cases but misses a considerable number of actual positive cases. Improving sensitivity is important in medical diagnosis.

Q4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs.

Answer:

i) Identify the epoch at which the model begins to overfit.

The model begins overfitting at Epoch 4, because validation loss starts increasing while training loss continues decreasing.

ii) Suggest two effective techniques to reduce overfitting.

1. Dropout: Randomly disables selected neurons during training to prevent excessive dependence on particular neurons.
2. Early Stopping: Stops training when validation loss begins increasing to prevent unnecessary learning from training data.

iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Overfitting occurs when a model learns training data too closely, including noise and unnecessary patterns. Consequently, training performance improves while performance on unseen data decreases, reducing the model's generalization ability.

Q5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs.

Answer:

i) Identify the epoch at which the model starts overfitting.

Overfitting becomes evident at Epoch 6, when validation accuracy decreases from 73% to 72% while training accuracy continues increasing.

ii) Explain why validation accuracy decreases despite training accuracy increasing.

The model increasingly memorizes the training data rather than learning general patterns. As a result, it performs poorly on unseen validation samples, causing validation accuracy to decrease.

iii) Suggest two methods to improve validation performance.

1. Data Augmentation: Increases training data variety through transformations such as rotation, flipping, and cropping.
2. Regularization: Techniques such as L1 and L2 regularization reduce excessive model complexity and improve generalization.

Q6. A machine-learning model shows the following loss values during training.

Answer:

i) At which epoch does the model achieve optimal generalization?

The model achieves optimal generalization at Epoch 4, where validation loss reaches its minimum value of 2.0.

ii) Identify signs of overfitting from the data.

- Training loss continuously decreases.
- Validation loss decreases until Epoch 4.
- Validation loss increases from Epoch 5 onward.
- The increasing gap between training and validation loss indicates overfitting.

iii) Recommend two regularization techniques and explain their impact.

1. L2 Regularization: Adds a penalty for large model weights, reducing model complexity and preventing overfitting.
2. Dropout: Randomly deactivates neurons during training, reducing dependence on specific neurons and improving generalization.

Q7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. The number of tires is 250 (125 front and 125 back tires). Front identified as Front: 119, Front identified as Back: 6, Back identified as Back: 120, Back identified as Front: 5. Compute Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Answer:

TP = 119, FN = 6, TN = 120, FP = 5

Accuracy

= (TP + TN) / (TP + TN + FP + FN) × 100

= (119 + 120) / 250 × 100

= 95.60%

Precision

= TP / (TP + FP) × 100

= 119 / 124 × 100

= 95.97%

Sensitivity (Recall)

= TP / (TP + FN) × 100

= 119 / 125 × 100

= 95.20%

Specificity

= TN / (TN + FP) × 100

= 120 / 125 × 100

= 96%

F1-Score

= 2TP / (2TP + FP + FN) × 100

= 238 / 249 × 100

= 95.58%

Q8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): Diseased identified as Diseased: 138, Diseased identified as Healthy: 12, Healthy identified as Healthy: 135, Healthy identified as Diseased: 15. Compute Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

Answer:

TP = 138, FN = 12, TN = 135, FP = 15

Accuracy

= (138 + 135) / 300 × 100

= 91%

Precision

= 138 / (138 + 15) × 100

= 90.20%

Sensitivity (Recall)

= 138 / (138 + 12) × 100

= 92%

Specificity

= 135 / (135 + 15) × 100

= 90%

F1-Score

= 2 × 138 / (2 × 138 + 15 + 12) × 100

= 276 / 303 × 100

= 91.09%

Q9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): Spam identified as

Spam: 180, Spam identified as Not Spam: 20, Not Spam identified as Not Spam: 170, Not Spam identified as Spam: 30. Compute Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Answer:

TP = 180, FN = 20, TN = 170, FP = 30

Accuracy

= (180 + 170) / 400 × 100

= 87.50%

Precision

= 180 / (180 + 30) × 100

= 85.71%

Sensitivity (Recall)

= 180 / (180 + 20) × 100

= 90%

Specificity

= 170 / (170 + 30) × 100

= 85%

F1-Score

= 2 × 180 / (2 × 180 + 30 + 20) × 100

= 360 / 410 × 100

= 87.80%

Q10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective): Defective identified as Defective: 230, Defective identified as Non-Defective: 20, Non-Defective identified as Non-Defective: 225, Non-Defective identified as Defective: 25. Compute Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Answer:

TP = 230, FN = 20, TN = 225, FP = 25

Accuracy

= (230 + 225) / 500 × 100

= 91%

Precision

= 230 / (230 + 25) × 100

= 90.20%

Sensitivity

= 230 / (230 + 20) × 100

= 92%

Specificity

= 225 / (225 + 25) × 100

= 90%

F1-Score

= 2 × 230 / (2 × 230 + 25 + 20) × 100

= 460 / 505 × 100

= 91.09%

Q11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated.

Answer:

P(X) = 1/3

P(Y) = 2/3

P(Adulterated|X) = 30/50 = 0.6

P(Adulterated|Y) = 20/60 = 1/3

Using Bayes' Theorem,

P(X|Adulterated) = [0.6 × (1/3)] / [(0.6 × 1/3) + (1/3 × 2/3)]

= 9/19

= 47.37%

P(Y|Adulterated) = [(1/3) × (2/3)] / [(0.6 × 1/3) + (1/3 × 2/3)]

= 10/19

= 52.63%

Q12. A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective.

Answer:

P(A) = 3/4

P(B) = 1/4

P(Defective|A) = 15/50 = 0.3

P(Defective|B) = 25/50 = 0.5

Using Bayes' Theorem,

P(A|Defective) = (0.3 × 0.75) / [(0.3 × 0.75) + (0.5 × 0.25)]

= 9/14

= 64.29%

P(B|Defective) = (0.5 × 0.25) / [(0.3 × 0.75) + (0.5 × 0.25)]

= 5/14

= 35.71%

Q13. In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces TP = 120 and TN = 150. The dataset contains 1000 samples per class, and 20% of the data is used for testing. Find Accuracy, Precision, Sensitivity (Recall), and Specificity.

Answer:

Total Samples = 1000 × 2 = 2000

Testing Samples = 2000 × 20/100 = 400

Positive Test Samples = 200

Negative Test Samples = 200

TP = 120, TN = 150

FN = 200 − 120 = 80

FP = 200 − 150 = 50

Accuracy

= (120 + 150) / 400 × 100

= 67.50%

Precision

= 120 / (120 + 50) × 100

= 70.59%

Sensitivity (Recall)

= 120 / (120 + 80) × 100

= 60%

Specificity

= 150 / (150 + 50) × 100

= 75%

Q14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data. Calculate Accuracy, Precision, Sensitivity, and Specificity.

Answer:

Total Samples = 1200 × 2 = 2400

Testing Samples = 2400 × 25/100 = 600

Positive Test Samples = 300

Negative Test Samples = 300

TP = 180, TN = 160

FN = 300 − 180 = 120

FP = 300 − 160 = 140

Accuracy

= (180 + 160) / 600 × 100

= 56.67%

Precision

= 180 / (180 + 140) × 100

= 56.25%

Sensitivity (Recall)

= 180 / (180 + 120) × 100

= 60%

Specificity

= 160 / (160 + 140) × 100

= 53.33%

Q15. A deep learning model is trained for a classification task, and the following loss values are observed. Answer the following: (a) At which epoch does the model achieve the best generalization performance? (b) At which point does overfitting begin? Justify your answer. (c) Suggest two techniques to reduce overfitting and briefly explain how they help.

Answer:

a. At which epoch does the model achieve the best generalization performance?

The model achieves the best generalization performance at Epoch 4, where validation loss reaches its minimum value of 1.8.

b. At which point does overfitting begin? Justify your answer.

Overfitting begins at Epoch 5, because training loss continues decreasing while validation loss increases from 1.8 to 1.9. This indicates that the model is learning the training data but its performance on unseen data is deteriorating.

c. Suggest two techniques to reduce overfitting and briefly explain how they help.

1. Early Stopping: Monitors validation loss and stops training when validation performance begins deteriorating, preventing unnecessary training.
2. Dropout: Randomly disables selected neurons during training, reducing dependence on specific neurons and improving generalization.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand